Microeconomics

Lecture 2 · The consumer problem

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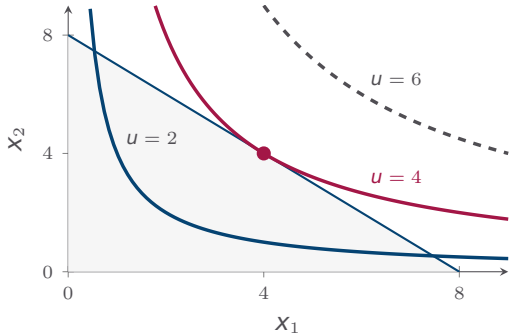
Artificial Intelligence and Financial Technologies

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From preferences to an actual choice

- Budget sets, existence and Marshallian demand.
- The Lagrangian, saddle points and Kuhn–Tucker conditions, with two proofs.
- Budget exhaustion and worked examples; then indirect utility, the envelope theorem and Roy’s identity.

The picture we already know



$u(x) = \sqrt{x_1 x_2}$.

$p = (1, 1)$, $m = 8$.

Which affordable bundle reaches the highest indifference curve?

Best bundle: $(4, 4)$.

Utility is 4.

The consumer problem

$$\begin{aligned} \max_{x} \quad & u(x) \\ \text{subject to} \quad & p \cdot x \leq m, \\ & x_i \geq 0 \quad \text{for every } i. \end{aligned}$$

Here $p \gg 0$ and $m > 0$. The feasible set is

$$B(p, m) = \{x \geq 0 : p \cdot x \leq m\}.$$

What did we assume when we wrote that?

- **Price-taking:** prices are given rather than negotiated.
- **Linear prices:** every unit has the same unit price.
- **Divisibility:** quantities can vary continuously.
- **Decision horizon:** include saving or future consumption when relevant.

What changes for a subscription, a quantity discount or an indivisible purchase?

Currency invariance of the budget

Claim. For every $t > 0$,

$$B(tp, tm) = B(p, m).$$

Proof. For any $x \geq 0$,

$$tp \cdot x \leq tm \quad \Longleftrightarrow \quad p \cdot x \leq m.$$

The same bundles are affordable. Only the currency units have changed.

The budget is nonempty and compact

Claim. If $p \gg 0$ and $m > 0$, then $B(p, m)$ is nonempty and compact.

Proof: nonempty. The zero bundle is affordable.

Closed. The inequalities $x_i \geq 0$ and $p \cdot x \leq m$ define closed sets.

Bounded. Every affordable bundle satisfies $0 \leq x_i \leq m/p_i$ for every i.

A closed and bounded subset of finite-dimensional Euclidean space is compact.

Does an optimum exist?

Assumptions: u is continuous on $\mathbb{R}^L_+$, $p \gg 0$ and $m > 0$.

The budget is nonempty and compact.

By the extreme value theorem, a continuous function attains a maximum on it.

Existence does not yet imply uniqueness, an interior solution or differentiability of demand.

Which assumption is doing the work?

Suppose $p_1 = 0$, $p_2 = 1$, $m = 1$, and $u(x) = x_1$.

Every quantity of good 1 is affordable. Utility is unbounded above, so there is no maximiser.

Zero prices do not always destroy existence, but compactness is no longer guaranteed.

Can you give a case with multiple maximisers even though the budget is compact?

For $u = x_1 + x_2$, $p = (1, 1)$ and $m = 1$, every bundle on $x_1 + x_2 = 1$ is optimal.

Marshallian demand is initially a correspondence

$$D(p, m) = \underset{x \geq 0,\, p \cdot x \leq m}{\arg\max}\, u(x).$$

$D(p, m)$ contains **all** optimal bundles. It may contain more than one.

When it is a singleton, write its unique element as $x(p, m)$: a demand function.

It is the earlier choice rule applied specifically to budget menus: $D(p, m) = C(B(p, m))$.

Proof: demand is homogeneous of degree zero

Claim: $D(tp, tm) = D(p, m)$ for every $t > 0$.

$tp \cdot x \le tm \iff p \cdot x \le m$: divide by the positive number t.

Therefore $B(tp, tm) = B(p, m)$. The objective u is the same.

The full argmax sets coincide. Neither uniqueness nor differentiability is needed.

Example: $(p, m) = ((2, 1), 12)$ and $((200, 100), 1200)$.

Let a multiplier enforce the budget

Keep the consumption domain $x \in X = \mathbb{R}^L_+$ for now; relax only $m - p \cdot x \geq 0$.

$$\ell(x, \lambda) = u(x) + \lambda(m - p \cdot x), \qquad \lambda \geq 0.$$

For an affordable bundle, $\ell(x, \lambda) \geq u(x)$. For an unaffordable one, increasing λ drives ℓ down.

One variable chooses consumption; the other enforces the budget. Their directions of optimisation differ.

Why minimise in the multiplier?

Fix a bundle $x \geq 0$ and minimise $\ell(x, \lambda)$ over $\lambda \geq 0$.

If $p \cdot x \leq m$, the minimum is $u(x)$, attained at $\lambda = 0$.

If $p \cdot x > m$, then $\inf_{\lambda \geq 0} \ell(x, \lambda) = -\infty$.

Thus the original problem can be written $\max_{x \geq 0} \inf_{\lambda \geq 0} \ell(x, \lambda)$.

This identity does not authorise exchanging max and inf.

A global saddle point

A pair $(x^*, \lambda^*) \in \mathbb{R}^L_+ \times \mathbb{R}_+$ is a global saddle if, for all $x \geq 0$ and $\lambda \geq 0$,

$$\ell(x, \lambda^*) \leq \ell(x^*, \lambda^*) \leq \ell(x^*, \lambda).$$

Left inequality: x^* maximises ℓ over the consumption domain, holding λ^* fixed.

Right inequality: λ^* minimises ℓ over nonnegative multipliers, holding x^* fixed.

Proof: the saddle enforces feasibility and slackness

Write $s = m - p \cdot x^*$. The right saddle inequality holds for every $\lambda \geq 0$.

If $s < 0$, then $\ell(x^*, \lambda) = u(x^*) + \lambda s \to -\infty$ as $\lambda \to \infty$, contradicting that inequality.

Hence $s \geq 0$: the budget is feasible. The condition $x^* \geq 0$ is already in the saddle domain.

Set $\lambda = 0$: $\lambda^* s \leq 0$. Nonnegative λ^* and s also give $\lambda^* s \geq 0$.

Therefore $\lambda^*(m - p \cdot x^*) = 0$ and $\ell(x^*, \lambda^*) = u(x^*)$.

Proof: a global saddle gives an optimal bundle

For every feasible y, nonnegative λ^* gives $u(y) \le \ell(y, \lambda^*)$.

The left saddle inequality gives $\ell(y, \lambda^*) \le \ell(x^*, \lambda^*)$.

Slackness completes the chain:

$$u(y) \le \ell(y, \lambda^*) \le \ell(x^*, \lambda^*) = u(x^*).$$

This direction needs no concavity or differentiability: the saddle inequalities are already global.

How could we find or verify a saddle?

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Minimisation in λ has given us feasibility and complementary slackness.

Maximisation in $x \geq 0$ remains: how do derivatives differ at positive and zero quantities?

If u is concave, then $\ell(\cdot, \lambda)$ is concave. A suitable first-order argument can certify its global maximum.

We now make the bounds $x_i \geq 0$ explicit using multipliers μ_i. → KKT back to a saddle

Why compare marginal utility per unit of money?

Shift a small amount of spending ε from good j to good i.

$$\Delta x_i = \varepsilon / p_i, \qquad \Delta x_j = -\varepsilon / p_j.$$

The first-order utility change is $\varepsilon\left(u_i/p_i - u_j/p_j\right)$.

If both goods are consumed in a smooth optimum, these marginal returns must agree.

The Lagrangian: one multiplier per inequality

Write the constraints as $m - p \cdot x \geq 0$ and $x_i \geq 0$.

$$\mathcal{L}(x, \lambda, \mu) = u(x) + \lambda(m - p \cdot x) + \sum_i \mu_i x_i.$$

The multipliers satisfy $\lambda \geq 0$, $\mu_i \geq 0$.

For the smooth argument, assume u is differentiable on an open neighbourhood of the candidate.

Stationarity in quantities

At a smooth candidate,

$$u_i(x^*) - \lambda^* p_i + \mu_i^* = 0 \qquad \text{for each } i.$$

One common budget multiplier, but a separate multiplier for every nonnegativity bound.

Stationarity is only one group of conditions. We must also check signs, feasibility and slackness.

The four Kuhn–Tucker groups

Stationarity: $u_i(x^*) - \lambda^* p_i + \mu_i^* = 0.$

Primal feasibility: $p \cdot x^* \leq m$, $x^* \geq 0.$

Dual feasibility: $\lambda^* \geq 0$, $\mu^* \geq 0.$

Complementary slackness:

$$\lambda^*(m - p \cdot x^*) = 0, \qquad \mu_i^* x_i^* = 0.$$

Complementary slackness: read the products

If the budget is slack, $m - p \cdot x^* > 0$, then $\lambda^* = 0$.

If $x_i^* > 0$, then $\mu_i^* = 0$.

If a multiplier is strictly positive, its constraint must bind.

Does a binding constraint always have a strictly positive multiplier?

For consumed goods: equal marginal returns

If $x_i^* > 0$, slackness gives $\mu_i^* = 0$, hence $u_i(x^*) = \lambda^* p_i$.

For two consumed goods with $u_j > 0$,

$$\mathrm{MRS}_{ij} = \frac{u_i(x^*)}{u_j(x^*)} = \frac{p_i}{p_j}.$$

The subjective marginal exchange rate equals the market exchange rate.

This is an interior condition, not a universal rule for every optimum.

For a good at zero: an inequality

If $x_i^* = 0$, stationarity and $\mu_i^* \geq 0$ imply

$$u_i(x^*) \leq \lambda^* p_i.$$

A consumed good has $u_i/p_i = \lambda$; an unconsumed good has $u_i/p_i \leq \lambda$.

At a corner, the unavailable direction of selling a zero holding matters.

Why would a rule that sets every marginal return equal miss some solutions?

A candidate is not yet a solution

Necessity: a differentiable local optimum satisfies KKT under a constraint qualification.

Here $p \gg 0$, $m > 0$ make the active linear constraints regular, including at corners. → Why?

Sufficiency: if u is differentiable and concave, a feasible KKT point is a global optimum.

Quasiconcavity alone does not make every stationary point optimal. → Sufficiency proof → Counterexample

Theorem 1: KKT is necessary

Let $p \gg 0$, $m > 0$, and let u be continuously differentiable on an open neighbourhood of the budget set.

If x^* is a local solution of the consumer problem, then there exist $\lambda^* \geq 0$, $\mu^* \geq 0$ satisfying all four KKT groups.

This includes corner optima and does not require concavity.

We prove this directly: construct the multipliers, prove their signs, then verify complementary slackness.

Construct the budget multiplier from an optimum

Take a nonzero optimum x^*; write $u_i = \partial u / \partial x_i$. Choose j with $x_j^* > 0$.

Define

$$\lambda^* := \frac{u_j(x^*)}{p_j}.$$

Feasible spending transfers between two consumed goods show that $u_i(x^*)/p_i = \lambda^*$ whenever $x_i^* > 0$.

Also $\lambda^* \geq 0$: otherwise $u_j(x^*) < 0$, and consuming slightly less of good j would be affordable and strictly improve utility.

Proof 1: consumed goods have one common return

If $x_i^* > 0$ and $x_j^* > 0$, a small spending transfer is possible in either direction.

If $u_i(x^*)/p_i > u_j(x^*)/p_j$, transfer ε of spending from j to i.

The utility derivative is $u_i/p_i - u_j/p_j > 0$: a feasible local improvement, contradicting optimality.

The reverse strict inequality is ruled out by the reverse transfer. Thus $u_i(x^*)/p_i = u_j(x^*)/p_j = \lambda^*$.

Construct the quantity-bound multipliers

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For each good, define

$$\mu_i^* := \begin{cases} 0, & x_i^* > 0, \\ \lambda^* p_i - u_i(x^*), & x_i^* = 0. \end{cases}$$

This definition makes stationarity hold:

$$u_i(x^*) - \lambda^* p_i + \mu_i^* = 0.$$

It also gives $\mu_i^* x_i^* = 0$.

The remaining question: why are these constructed μ_i^* nonnegative? We have not assumed their sign.

If a constructed multiplier were negative

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Only $x_i^* = 0$ needs checking: for a consumed good we already set $\mu_i^* = 0$.

Suppose $\mu_i^* < 0$. By its definition,

$$\lambda^* p_i - u_i(x^*) < 0 \quad \Rightarrow \quad \lambda^* < \frac{u_i(x^*)}{p_i}.$$

For the consumed good j, $\lambda^* = u_j(x^*)/p_j$. Therefore

$$\frac{u_i(x^*)}{p_i} > \frac{u_j(x^*)}{p_j}.$$

The unconsumed good would give more marginal utility per unit of spending.

Proof: a profitable spending transfer

Shift spending ε from consumed good j to unconsumed good i:

$$x_i(\varepsilon) = \frac{\varepsilon}{p_i}, \qquad x_j(\varepsilon) = x_j^* - \frac{\varepsilon}{p_j}.$$

Keep other quantities fixed. For $0 < \varepsilon < p_j x_j^*$, quantities remain nonnegative and total spending is unchanged.

$$\frac{d}{d\varepsilon} u(x(\varepsilon))\bigg|_{0+} = \frac{u_i(x^*)}{p_i} - \frac{u_j(x^*)}{p_j} = -\frac{\mu_i^*}{p_i} > 0.$$

A sufficiently small transfer strictly improves utility: contradiction. Thus $\mu_i^* \geq 0$, with equality allowed.

Proof 1: finish with complementary slackness

If $p \cdot x^* = m$, then $\lambda^*(m - p \cdot x^*) = 0$.

If the budget is slack and $x_j^* > 0$, both a small increase and a small decrease in x_j are feasible.

Local optimality forces $u_j(x^*) = 0$, hence $\lambda^* = u_j(x^*)/p_j = 0$. The budget product is again zero.

For every i, the construction gives $\mu_i^* x_i^* = 0$: either the quantity or its multiplier is zero.

Feasibility came from x^*; stationarity, multiplier signs and slackness have now all been proved.

Proof 1: the zero-bundle case

If $x^* = 0$, we cannot select a consumed good to define λ^*.

Because $m > 0$ and $p \gg 0$, a sufficiently small increase in any one good is affordable.

Local optimality gives $u_i(0) \leq 0$ for every i.

Choose $\lambda^* = 0$ and $\mu_i^* = -u_i(0) \geq 0$. Stationarity and both complementary-slackness products hold.

This completes the necessity proof, including all corners and the zero bundle.

Theorem 2: concave KKT is sufficient

Let u be concave on $\mathbb{R}^L_+$ and differentiable on a neighbourhood of a candidate x^*.

If (x^*, λ^*, μ^*) satisfies all four KKT groups, then x^* is a global solution of the consumer problem.

We will first maximise the Lagrangian at fixed multipliers, then use feasibility and complementary slackness.

The global argument uses concavity of the chosen index, not merely convexity of preferences.

Fix the multipliers and look only at quantities

Define $f(x) = \mathcal{L}(x, \lambda^*, \mu^*)$, holding the multipliers fixed.

$$f(x) = u(x) + \lambda^* m - \lambda^* p \cdot x + \mu^* \cdot x.$$

If u is concave, then f is concave: the added terms are affine in x.

The constructed stationarity conditions give $\nabla f(x^*) = 0$. Why is that a global maximum?

Proof: a stationary point of a concave function

A differentiable concave function lies below its tangent plane. For every y in its convex domain,

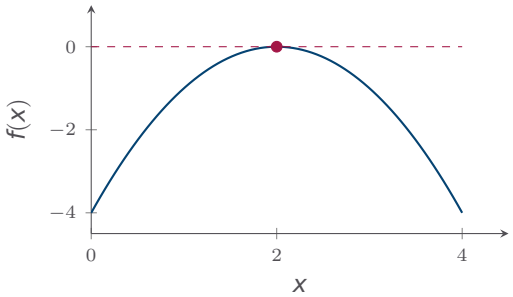
$$f(y) \leq f(x^*) + \nabla f(x^*) \cdot (y - x^*).$$

With $\nabla f(x^*) = 0$, this becomes

$$f(y) \leq f(x^*).$$

Thus x^* globally maximises $\mathcal{L}(\cdot, \lambda^*, \mu^*)$ on that domain. Uniqueness is not asserted.

A horizontal tangent above the whole graph



$f(x) = -(x-2)^2$.
The derivative is zero at $x = 2$.
The horizontal tangent is an upper bound on the entire graph.
Zero gradient means a flat tangent, not a constant function nearby.

Proof 2: return from the Lagrangian to utility

For any feasible y, nonnegative multipliers give

$$u(y) \leq \mathcal{L}(y, \lambda^*, \mu^*).$$

Concavity and stationarity have shown

$$\mathcal{L}(y, \lambda^*, \mu^*) \leq \mathcal{L}(x^*, \lambda^*, \mu^*).$$

Complementary slackness gives $\mathcal{L}(x^*, \lambda^*, \mu^*) = u(x^*)$. Thus $u(y) \leq u(x^*)$ for every feasible y.

The same slackness and feasibility minimise $\mathcal{L}(x^*, \lambda, \mu)$ over nonnegative multipliers: the saddle interpretation is recovered.

Budget exhaustion: Walras' law

If preferences are locally nonsatiated, every $x^* \in D(p, m)$ satisfies $p \cdot x^* = m$.

Assume an optimum exists. What would happen if it left strictly positive budget slack?

The key: with slack, every sufficiently small change in the consumption domain remains affordable.

Proof: LNS and budget exhaustion

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Suppose an optimum x^* has slack $\delta = m - p \cdot x^* > 0$.

If $\|y - x^*\| < \delta / \|p\|$, then

$$p \cdot y \leq p \cdot x^* + \|p\| \|y - x^*\| < m.$$

LNS supplies a strictly better such y in the consumption domain, contradicting optimality.

→ Back to budget exhaustion

Proof: convex preferences give a convex demand set

Assume preferences are convex and take $x, y \in D(p, m)$.

For $t \in [0, 1]$, $z_t = tx + (1 - t)y$ is affordable because $B(p, m)$ is convex.

For every affordable w, optimality gives $x \succeq w$ and $y \succeq w$.

Convex preferences imply $z_t \succeq w$ for every such w. Thus $z_t \in D(p, m)$.

Hence $D(p, m)$ is convex. It can still be a whole line segment.

Strict convexity: at most one optimum

If preferences are strictly convex, there is at most one optimal bundle on a convex budget set.

Together with existence, this gives a demand function $x(p, m)$.

Does strict convexity alone establish existence?

No. Continuity and compactness gave existence; strict convexity gives at most one optimum.

Proof: uniqueness from strict convexity

Suppose distinct $x, y \in D(p, m)$. Their midpoint z is feasible because the budget is convex.

Optimality gives $x \sim y$. Strict convexity of preferences implies $z \succ x$.

This contradicts x being optimal. There is at most one optimum; existence gives a unique one.

→ Back to uniqueness

Before differentiating, inspect the economics

Substitutes: compare utility gained per unit of spending; a corner may be optimal.

Complements: look for useful proportions; the optimum may be a kink.

Smooth interior problem: marginal conditions can identify a candidate.

Always check feasibility, boundary cases and a reason why the candidate is globally optimal.

Checkpoint

What established existence? What could establish uniqueness?

Which KKT condition handles a zero quantity?

When is a KKT solution sufficient for global optimality?

What can go wrong if we check only the interior first-order equations?

Example 1: Cobb–Douglas

$$u(x) = x_1^{\alpha} x_2^{1-\alpha}, \qquad 0 < \alpha < 1.$$

Could an optimum lie on an axis? Which representation would make the calculation easier?

An axis bundle gives zero utility; a strictly positive affordable bundle gives positive utility.

Every optimum is therefore positive. We may now take logs and use an interior method.

Choose a convenient representation

Maximising u is equivalent, at the positive optimum, to maximising

$$\widetilde{u}(x) = \alpha \log x_1 + (1-\alpha)\log x_2.$$

$$\mathcal{L} = \alpha \log x_1 + (1-\alpha)\log x_2 + \lambda(m - p_1 x_1 - p_2 x_2).$$

The log representation is concave on the positive domain.

The bundle does not change, but the numerical utility and multiplier generally do.

Solve the interior conditions

$$\frac{\alpha}{x_1} = \lambda p_1, \qquad \frac{1-\alpha}{x_2} = \lambda p_2.$$

Thus $p_1 x_1 = \alpha/\lambda$ and $p_2 x_2 = (1-\alpha)/\lambda$.

Budget exhaustion gives $m = 1/\lambda$, hence $\lambda = 1/m$.

$$x_1(p, m) = \frac{\alpha m}{p_1}, \qquad x_2(p, m) = \frac{(1-\alpha)m}{p_2}.$$

Interpret the solution

$$\frac{p_1 x_1}{m} = \alpha, \qquad \frac{p_2 x_2}{m} = 1 - \alpha.$$

Expenditure shares are constant in this specification.

Demand is homogeneous of degree zero and exhausts the budget.

If $\alpha = 1/2$, $p = (2, 1)$ and $m = 12$, demand is $(3, 6)$.

Which of these predictions is a modelling restriction rather than an accounting identity?

A price counterfactual

Keep $\alpha = 1/2$ and $m = 12$. Raise p_1 from 2 to 3, holding $p_2 = 1$.

What are the new quantities? The spending shares? The attained raw utility?

Does an unchanged quantity of good 2 mean that its utility contribution is unchanged?

Example 2: perfect substitutes

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$$u(x) = ax_1 + bx_2, \qquad a, b > 0.$$

Where would you spend the next unit of money? Can the optimum be nonunique?

Compare a/p_1 and b/p_2. Buy $(m/p_1, 0)$ if the first is larger; buy $(0, m/p_2)$ if the second is larger.

If the ratios are equal, every budget-line bundle is optimal.

Verify a corner with the full KKT system

Let $u(x) = 2x_1 + x_2$, $p = (3, 1)$ and $m = 12$.

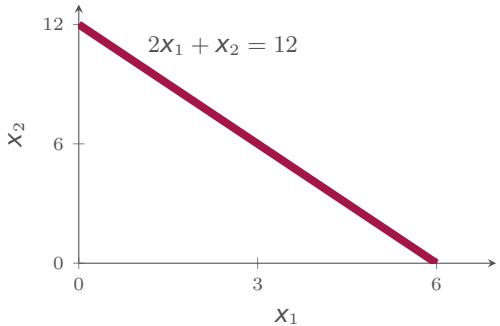
The spending returns are $2/3$ and 1, so $x^* = (0, 12)$.

Stationarity and slackness give $\lambda = 1$, $\mu_1 = 1$, $\mu_2 = 0$.

Check: $2 - 3\lambda + \mu_1 = 0$, $1 - \lambda + \mu_2 = 0$.

The zero quantity is optimal precisely because its marginal return is too low.

Equal returns: a whole set of demand



$u = 2x_1 + x_2$, $p = (2, 1)$, $m = 12$.
Every budget-line bundle gives utility 12.
$D(p, m)$ is a segment. Every optimal bundle gives utility 12.

Example 3: lessons and money left over

$$u(a,q) = a + 4\log(1+q), \qquad a + rq \le m, \quad a, q \ge 0.$$

The price of a is 1; the lesson price is $r > 0$; wealth is $m > 0$.

Must both goods be consumed? How could we reduce this to one variable?

Budget exhaustion gives $a = m - rq$ and $0 \le q \le m/r$.

Solve first, then respect the quantity bounds

$$\max_{0 \le q \le m/r} \; m - rq + 4\log(1+q).$$

The derivative $-r + 4/(1+q)$ is strictly decreasing.

The stationary quantity is $4/r - 1$. Impose both bounds:

$$q^* = \min\left\{\frac{m}{r}, \max\left\{\frac{4}{r} - 1, 0\right\}\right\}, \quad a^* = m - rq^*.$$

Three regimes, one decision problem

If $r \geq 4$: no tutoring, $q^* = 0$, $a^* = m$.

If $0 < r < 4$ and $m \leq 4 - r$: the no-borrowing bound binds, $q^* = m/r$, $a^* = 0$.

If $0 < r < 4$ and $m > 4 - r$: both goods are consumed, $q^* = 4/r - 1$, $a^* = m - 4 + r$.

At $r = 1$: income 2 gives $(a, q) = (0, 2)$; income 8 gives $(5, 3)$.

At $m = 4 - r$, both formulas agree; a binding bound may have zero multiplier.

Example 4: perfect complements

$$u(x) = \min\{x_1, x_2\}.$$

Would you pay for a useless surplus of one good? Can we differentiate at the candidate?

Utility t requires both quantities at least t. The cheapest attainment is (t, t), costing $(p_1 + p_2)t$.

The highest affordable target is $t = m/(p_1 + p_2)$.

A fixed-proportion solution

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$$x_1^* = x_2^* = \frac{m}{p_1 + p_2}.$$

The cheapest implementation of the highest target costs the entire budget. Any surplus would cost extra, so the optimum is unique.

At $p = (2, 1)$ and $m = 12$, demand is $(4, 4)$.

The optimum is a kink: the target-cost argument replaces a smooth tangency calculation.

A practical solution checklist

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1. **Diagnose:** domain, continuity, corners, kinks, satiation.
2. **Establish:** existence, and uniqueness if justified.
3. **Solve:** geometry, a useful substitution or the relevant KKT cases.
4. **Verify:** feasibility, multiplier signs, slackness and global optimality.
5. **Interpret:** quantities, expenditure shares and reactions to parameters.

From a classroom solution to a structural model

Observed $p = (2, 1)$, $m = 12$, $x = (3, 6)$ imply a spending share of $1/2$ on good 1.

Within a Cobb–Douglas specification, this suggests $\alpha = 1/2$.

With the same preference parameter, $p_1 = 3$ predicts $x = (2, 6)$.

Fitting one choice does not establish the functional form. Use more variation and test the restrictions.

Indirect utility: the optimal value

$$v(p, m) = \max_{x \geq 0,\, p \cdot x \leq m} u(x).$$

Demand tells us **which bundle**; indirect utility tells us **the attained value**.

Even if $D(p, m)$ contains several bundles, every one gives the same value $v(p, m)$.

It is a function of the environment, not of a chosen bundle.

Indirect utility: the properties to prove

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- Maintain continuous u, $p \gg 0$ and $m > 0$, so optimal values exist.
- $v(tp, tm) = v(p, m)$; higher prices cannot increase v.
- More wealth weakly increases v; under LNS the increase is strict.
- v is continuous on the positive-price, positive-wealth domain.

Proof: currency units and higher prices

$B(tp, tm) = B(p, m)$ and the same u is maximised, so $v(tp, tm) = v(p, m)$.

If $q \geq p$ componentwise and $x \geq 0$, then $p \cdot x \leq q \cdot x$.

Thus $q \cdot x \leq m$ implies $p \cdot x \leq m$: $B(q, m) \subseteq B(p, m)$.

Maximising the same utility over a smaller set cannot give a larger value: $v(q, m) \leq v(p, m)$.

Proof: more wealth and local nonsatiation

Let $m' > m$ and choose $x \in D(p, m)$.

Choose $\varepsilon = (m' - m)/(2\|p\|) > 0$.

By LNS, some $y \geq 0$ satisfies $\|y - x\| < \varepsilon$ and $u(y) > u(x) = v(p, m)$.

$p \cdot y \leq p \cdot x + \|p\|\|y - x\| < m + (m' - m)/2 < m'$.

So y is affordable at the new wealth and $v(p, m') \geq u(y) > v(p, m)$.

Continuity: put every budget on the same simplex

Define the fixed compact set $\Delta = \{z \geq 0 : \sum_i z_i \leq 1\}$.

For each budget, write $z_i = p_i x_i / m$, or equivalently $x_i = (m/p_i) z_i$.

This is a bijection between $B(p, m)$ and Δ, since $p \cdot x / m = \sum_i z_i$.

With $g(p, m, z) = u((m/p_1) z_1, \ldots, (m/p_L) z_L)$, we have $v(p, m) = \max_{z \in \Delta} g(p, m, z)$.

Proof: the optimal value is continuous

Take $(p^n, m^n) \to (p, m)$ with $p \gg 0$, $m > 0$.

Choose a compact parameter neighbourhood K containing the tail of this sequence, with all prices and wealth bounded away from zero.

g is continuous on the compact set $K \times \Delta$, hence uniformly continuous. Therefore $\sup_{z \in \Delta} |g(p^n, m^n, z) - g(p, m, z)| \to 0$.

$$|v(p^n, m^n) - v(p, m)| \leq \sup_{z \in \Delta} |g(p^n, m^n, z) - g(p, m, z)| \to 0.$$

No differentiability or uniqueness of the optimiser was needed.

A less obvious property: quasiconvexity

NES

For $0 \leq t \leq 1$, set $(p^t, m^t) = t(p, m) + (1 - t)(q, n)$.

$$v(p^t, m^t) \leq \max\{v(p, m), v(q, n)\}.$$

If a bundle is unaffordable at both endpoints, it is unaffordable at their mixture too.

So the mixed budget cannot deliver utility above both endpoint values. → Proof

Proof: indirect utility is quasiconvex

NES

Write $(p^t, m^t) = t(p, m) + (1 - t)(q, n)$ with $0 < t < 1$.

If x violates both budgets, then $p \cdot x > m$ and $q \cdot x > n$, hence $p^t \cdot x > m^t$.

Thus every bundle affordable at the mixture is affordable at at least one endpoint.

Its utility cannot exceed $\max\{v(p, m), v(q, n)\}$. Maximise over the mixed budget.

→ Back to the property

Compute indirect utility in the same example

NES

For the log representation, substitute the demands into $\widetilde{u}$:

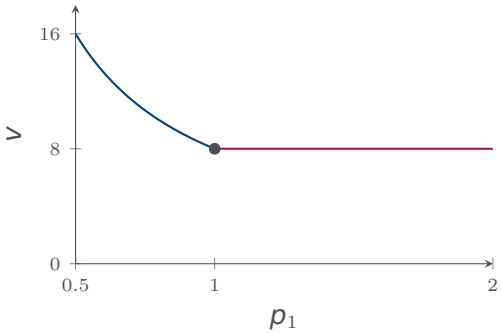
$$v(p, m) = \alpha \log \frac{\alpha m}{p_1} + (1 - \alpha) \log \frac{(1 - \alpha) m}{p_2}.$$

$$v(p, m) = \log m - \alpha \log p_1 - (1 - \alpha) \log p_2 + c_\alpha,$$

where $c_\alpha = \alpha \log \alpha + (1 - \alpha) \log(1 - \alpha)$.

→ More than two goods

A continuous value need not be differentiable



$u = x_1 + x_2$, $p_2 = 1$, $m = 8$.

$$v(p_1, 1, 8) = \frac{8}{\min\{p_1, 1\}}.$$

At $p_1 = 1$, the consumer is indifferent along the budget line. The value has a kink.

The envelope question

We know the optimal bundle and the attained value $v(p, m)$.

If a parameter changes, the optimal choice usually changes too.

Can we find the first-order value change without differentiating the optimal choice?

First a theorem for a general constrained problem; then its consumer application.

A general parameterised optimisation problem

$$V(\theta) = \max_{x} f(x, \theta) \quad \text{subject to} \quad g_j(x, \theta) \geq 0, \quad h_k(x, \theta) = 0.$$

The parameter varies in an open set; objective and constraints are continuously differentiable.

The ambient decision domain is open; all quantity bounds are included among the constraints.

Use $\lambda_j \geq 0$ for inequalities and unrestricted ν_k for equalities.

The general Lagrangian

$$\mathcal{L} = f(x, \theta) + \sum_j \lambda_j g_j(x, \theta) + \sum_k \nu_k h_k(x, \theta).$$

At an optimum: $\mathcal{L}_x = 0$, $g_j \geq 0$, $h_k = 0$, $\lambda_j \geq 0$ and $\lambda_j g_j = 0$.

Assume a continuously differentiable selection of optimal choices and KKT multipliers exists near the parameter under consideration.

This is a sufficient smooth-branch regularity assumption, to be checked in an application.

Envelope theorem: a constrained smooth version

Under the preceding smooth-branch assumptions,

$$\frac{\partial V}{\partial \theta_\ell} = \frac{\partial \mathcal{L}}{\partial \theta_\ell}(x^*, \lambda^*, \nu^*; \theta).$$

On the right, hold decisions and multipliers fixed when taking the partial derivative.

Thus $V_{\theta_\ell} = f_{\theta_\ell} + \sum_j \lambda_j^* (g_j)_{\theta_\ell} + \sum_k \nu_k^* (h_k)_{\theta_\ell}$, evaluated at the optimum.

Proof: differentiate along an optimal branch

Feasibility and slackness give $V(\theta) = \mathcal{L}(x^*(\theta), \lambda^*(\theta), \nu^*(\theta); \theta)$.

Fix one coordinate t of θ. The chain rule gives

$$\frac{dV}{dt} = \mathcal{L}_t + \mathcal{L}_x x_t^* + \sum_j g_j(\lambda_j^*)_t + \sum_k h_k(\nu_k^*)_t.$$

Stationarity kills the $\mathcal{L}_x$ term; equality feasibility kills the h_k terms.

Proof: why multiplier derivatives disappear

NES

For each inequality, examine $g_j(\lambda_j^*)_t$.

If $\lambda_j^* > 0$, complementary slackness gives $g_j = 0$.

If $\lambda_j^* = 0$, its differentiable nonnegative branch has a local minimum at zero, so $(\lambda_j^*)_t = 0$.

Every extra term is zero. Hence

$$\frac{dV}{dt} = \mathcal{L}_t.$$

Apply the theorem to the consumer

Here $f = u(x)$ and the parameter-dependent constraint is $m - p \cdot x \geq 0$. Quantity bounds do not depend on prices or wealth.

Where the envelope assumptions hold,

$$v_m = \mathcal{L}_m = \lambda^*, \qquad v_{p_i} = \mathcal{L}_{p_i} = -\lambda^* x_i^*.$$

Wealth relaxes the budget; a higher price tightens it in proportion to the quantity purchased.

The budget multiplier has an interpretation

$$\lambda^* = v_m(p, m).$$

It is marginal utility from one extra unit of wealth: the value of relaxing the budget.

Its numerical size depends on the chosen utility scale.

A binding budget does not by itself imply $v_m > 0$. → A counterexample

Roy's identity: recover demand from value

Assume continuous utility, LNS, $p \gg 0$, $m > 0$, differentiable v and $v_m > 0$.

Every optimiser satisfies

$$x_i^* = -\frac{v_{p_i}(p, m)}{v_m(p, m)}.$$

At a smooth regular optimum, the envelope formulas give the ratio by cancelling λ^*.

The next proof needs no differentiable demand or differentiable direct utility.

Roy without differentiating demand

Let $x^* \in D(p, m)$. LNS gives $p \cdot x^* = m$.

At any nearby price vector q, the bundle x^* is affordable with wealth $q \cdot x^*$. Thus $F(q) = v(q, q \cdot x^*) \geq u(x^*) = F(p)$.

If v is differentiable, F has a minimum at p:

$$0 = F_{q_i}(p) = v_{p_i}(p, m) + v_m(p, m) x_i^*.$$

If $v_m > 0$, every optimiser satisfies the same Roy formula, so demand is unique.

→ Back

Check Roy in the Cobb–Douglas example

$$v = \log m - \alpha \log p_1 - (1-\alpha)\log p_2 + c_\alpha.$$

$$v_m = 1/m, \qquad v_{p_1} = -\alpha/p_1, \qquad v_{p_2} = -(1-\alpha)/p_2.$$

$$x_1 = -\frac{-\alpha/p_1}{1/m} = \frac{\alpha m}{p_1}.$$

Likewise $x_2 = (1-\alpha)m/p_2$. The demand agrees with the direct solution.

Roy example 2: substitutes and a corner

For $u = ax_1 + bx_2$,

$$v(p, m) = m \max\left\{\frac{a}{p_1}, \frac{b}{p_2}\right\}.$$

Differentiate in a region where good 1 gives the strictly larger spending return.

If $a/p_1 > b/p_2$, then $v = am/p_1$, $v_m = a/p_1$, $v_{p_1} = -am/p_1^2$ and $v_{p_2} = 0$.

Roy gives $x_1 = m/p_1$, $x_2 = 0$.

The other corner and the tie

If $a/p_1 < b/p_2$, the calculation gives $x_1 = 0$, $x_2 = m/p_2$.

For $u = 2x_1 + x_2$, $p = (3, 1)$, $m = 12$, Roy recovers $(0, 12)$.

If the returns are equal, the value has a price kink: the active formulas have different price gradients.

At the tie Roy's differentiability assumption fails; demand is a whole segment.

Roy example 3: the logarithmic model

Return to $u = a + 4\log(1+q)$ and $a + rq \le m$. Substitute the demands:

$$v(r,m) = \begin{cases} m, & r \ge 4, \\ 4\log(1+m/r), & 0 < r < 4,\ m \le 4 - r, \\ m - 4 + r + 4\log(4/r), & 0 < r < 4,\ m > 4 - r. \end{cases}$$

Can the same derivative ratio recover all three demand regimes?

One ratio, three regimes

Compute $q^* = -v_r/v_m$ in each region.

Regime	v_m	v_r	$-v_r/v_m$
No tutoring	1	0	0
All wealth on q	$\frac{4}{r+m}$	$-\frac{4m}{r(r+m)}$	$\frac{m}{r}$
Both goods	1	$1-\frac{4}{r}$	$\frac{4}{r}-1$

Recover $a^* = m - rq^*$. At $m = 4 - r$ and at $r = 4$, adjacent values and first derivatives agree.

Roy example 4: a kink in direct utility

For $u = \min\{x_1, x_2\}$,

$$v(p, m) = \frac{m}{p_1 + p_2}.$$

$v_m = 1/(p_1 + p_2)$ and $v_{p_i} = -m/(p_1 + p_2)^2$.

$$x_i = -\frac{v_{p_i}}{v_m} = \frac{m}{p_1 + p_2} \qquad (i = 1, 2).$$

Direct utility has a kink, but indirect utility is differentiable. The direct proof of Roy still applies.

Demand survives a change of utility scale

If $\widehat{v} = f(v)$ with $f'(v) > 0$, then

$$\widehat{v}_{p_i} = f'(v) v_{p_i}, \qquad \widehat{v}_m = f'(v) v_m.$$

The common factor cancels from Roy's ratio.

The multiplier changes, but demand does not.

A strictly increasing transformation can have a zero derivative at a point; then the differentiable ratio may be unavailable.

Checkpoint

Which assumptions guarantee existence? Which give uniqueness?

Why can $D(p, m)$ be multivalued while $v(p, m)$ is single-valued?

What replaces equality of marginal returns for an unconsumed good?

What justifies treating a KKT candidate as globally optimal?

What two differentiability/positivity conditions are needed to use Roy's formula?

Read next, and keep the core route clear

Varian: *Intermediate Microeconomics: A Modern Approach*, 9th edition (2014), Chapter 5 *Choice* and Chapter 6 *Demand*.

Advanced: Mas-Colell, Whinston and Green (1995), Chapter 3, section 3.D *The Utility Maximization Problem*; the Mathematical Appendix on constrained optimisation.

The next course block develops expenditure minimisation, Hicksian demand and duality. Reserve slides follow.

Homework 1 covers Lectures 1–2 and is due on 14 November 2026.

Next block: two ways to ask the consumer question

Utility maximisation: given p, m, how much utility can the consumer attain?

Expenditure minimisation: given $p, \bar{u}$, how little spending attains the target?

One fixes resources; the other fixes a level of the same preferences.

Expenditure minimisation

$$e(p, \bar{u}) = \min_{x \geq 0,\, u(x) \geq \bar{u}} p \cdot x.$$

$$H(p, \bar{u}) = \arg\min_{x \geq 0,\, u(x) \geq \bar{u}} p \cdot x.$$

e is the expenditure function; H is the Hicksian demand correspondence.

Assume the target is attainable. A singleton is denoted $h(p, \bar{u})$.

Why an unbounded feasible set can still have a minimum

NES

Assume $p \gg 0$, continuous u, and pick a bundle $\widehat{x}$ with $u(\widehat{x}) \geq \bar{u}$.

Only consider target-feasible bundles with $p \cdot x \leq p \cdot \widehat{x}$.

This set is nonempty, closed and bounded, hence compact.

A continuous cost attains a minimum there; excluded bundles are more expensive. → Details

Hicksian demand: basic properties

Price homogeneity: $H(tp, \bar{u}) = H(p, \bar{u})$ for $t > 0$.

No excess utility: if $\bar{u} \geq u(0)$, every minimiser attains exactly $\bar{u}$.

Convex preferences give a convex minimiser set; strict convexity, with the maintained conditions, gives uniqueness.

No excess utility uses continuity and positive prices, not LNS.

The expenditure function has a different curvature

NES

Homogeneity of degree one: $e(tp, \bar{u}) = te(p, \bar{u})$.

It is nondecreasing in prices and, above $u(0)$ on attainable targets, strictly increasing in the target.

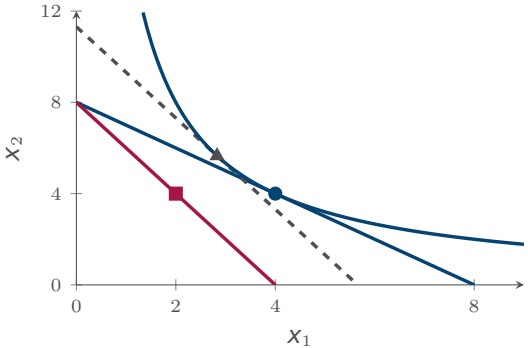
With continuous utility and LNS, expenditure is continuous on the positive-price, attainable-target domain.

Concavity in prices:

$$e(tp + (1-t)q, \bar{u}) \geq te(p, \bar{u}) + (1-t)e(q, \bar{u}).$$

It is the infimum of linear cost functions. → Proofs

Fixed wealth versus fixed utility



$u = \sqrt{x_1 x_2}$; initially $p = (1, 1)$, $m = 8$, $x = (4, 4)$.
Raise p_1 to 2: Marshallian demand at the same wealth is $(2, 4)$.
To keep utility 4, spend $8\sqrt{2}$ and buy $(2\sqrt{2}, 4\sqrt{2})$.

Two identities to develop next time

Under continuous LNS preferences, positive prices and an attainable target $\bar{u} > u(0)$:

$$e\big(p, v(p,m)\big) = m, \qquad v\big(p, e(p,\bar{u})\big) = \bar{u}.$$

For unique demands,

$$h\big(p, v(p,m)\big) = x(p,m).$$

These connect the two problems; they are not an assumption that the consumer literally solves both.

A monetary compensation question

Let $u^0 = v(p^0, m)$. How much extra wealth restores this utility after a price change to p^1?

$$\text{CV} = e(p^1, u^0) - m.$$

In the diagram, $\text{CV} = 8\sqrt{2} - 8 \approx 3.31$.

A money-metric change is useful, but summing it across people still involves welfare judgements.

Why the constraint qualification holds here

Active nonnegativity constraints have gradients e_i for the coordinates at zero.

If the budget is slack, these vectors are linearly independent.

If the budget binds, $m > 0$ implies some $x_j > 0$. The budget gradient $-p$ has a nonzero j coordinate, unlike every active bound gradient.

The active gradients are therefore linearly independent. This includes corner optima.

→ Back to necessity and sufficiency

Proof: why concavity makes KKT sufficient

NES

For feasible y, concavity and differentiability imply

$$u(y) - u(x^*) \leq \nabla u(x^*) \cdot (y - x^*).$$

Stationarity gives $\nabla u(x^*) = \lambda^* p - \mu^*$.

Budget feasibility and slackness give $\lambda^* p \cdot (y - x^*) \leq 0$. Bound feasibility and slackness give $-\mu^* \cdot (y - x^*) \leq 0$.

Thus $u(y) \leq u(x^*)$ for every feasible y.

→ Back

Proof: concave KKT gives the budget saddle

Let u be concave and differentiable and let (x^*, λ^*, μ^*) satisfy KKT.

For $y \geq 0$, concavity gives $\ell(y, \lambda^*) - \ell(x^*, \lambda^*) \leq (\nabla u(x^*) - \lambda^* p) \cdot (y - x^*)$.

Stationarity makes this $-\mu^* \cdot (y - x^*) = -\mu^* \cdot y \leq 0$, using $\mu^* \cdot x^* = 0$.

Budget feasibility and slackness give $\ell(x^*, \lambda^*) = u(x^*) \leq \ell(x^*, \lambda)$ for every $\lambda \geq 0$.

Both saddle inequalities hold. → Back to the bridge

A KKT point that is not optimal

Let $u(x) = (x-1)^3$, $p = 1$, $m = 2$, $x \geq 0$.

At $x = 1$, take $\lambda = \mu = 0$. Stationarity, feasibility and complementary slackness all hold.

But $u(2) = 1 > 0 = u(1)$. The KKT point is not optimal.

The function is strictly increasing and quasiconcave. That alone does not make stationarity sufficient.

→ Back

A binding budget with zero marginal value

NES

Use $u(x) = (x-1)^3$, but now $p = 1$, $m = 1$.

The unique optimum is $x^* = 1$: utility is strictly increasing. The budget binds.

Yet $u'(1) = 0$, $\lambda = 0$, and $v_m(1,1) = 0$.

Roy's ratio becomes $0/0$ at that point, although demand $x = m/p$ is perfectly well-defined.

→ Back to the multiplier

Cobb–Douglas with any positive exponent sum

Let $u(x) = \prod_{i=1}^{L} x_i^{a_i}$ with $a_i > 0$ and $A = \sum_i a_i$.

After proving interiority, maximise $\sum_i a_i \log x_i$. Stationarity gives $x_i = a_i/(\lambda p_i)$.

Budget exhaustion gives $\lambda = A/m$, hence

$$x_i(p, m) = \frac{a_i}{A} \frac{m}{p_i}.$$

No restriction $A \leq 1$ is needed for this preference solution.

→ Back

Proof: expenditure minimisation has a solution

Choose an attainable target and a feasible $\widehat{x}$. Let

$$S = \{x \geq 0 : u(x) \geq \bar{u}\} \cap \{x : p \cdot x \leq p \cdot \widehat{x}\}.$$

S is nonempty and compact when u is continuous and $p \gg 0$.

Cost is continuous, so it has a minimum on S. Every omitted feasible bundle costs more than $\widehat{x}$, so this is a global minimum.

→ Back to the preview

Proof: homogeneity and price concavity of expenditure

Scaling prices multiplies every feasible cost by the same positive number:

$$e(tp, \bar{u}) = te(p, \bar{u}), \qquad H(tp, \bar{u}) = H(p, \bar{u}).$$

Let x^t minimise cost at $p^t = tp + (1-t)q$. It is feasible for the target at either price vector.

$$e(p^t, \bar{u}) = tp \cdot x^t + (1-t)q \cdot x^t \geq te(p, \bar{u}) + (1-t)e(q, \bar{u}).$$

→ Back to expenditure properties